Categorical Variables

Dummies, baseline/offset form, and interactions

Foundations of Regression and Predictive Modeling

James Scott | Abhra Sarkar | UT-Austin

Grouped data

Imagine a streaming service testing three cancellation-page offers:

- no offer,
- one free month,
- switch to a cheaper annual plan.

Each row in the data is a customer who clicked "cancel."

The business question:
Which offer keeps more customers active 30 days later?

The motif

This one example of a common motif:

> *use regression to compare outcomes across named conditions.*

For example:

- late-delivery rates under different carrier contracts,
- readmission rates under different discharge protocols,
- support resolution times under different routing rules.

The trick is to write the comparison directly into the model. To do that, we need to talk about how we encode numbers.

Raw form

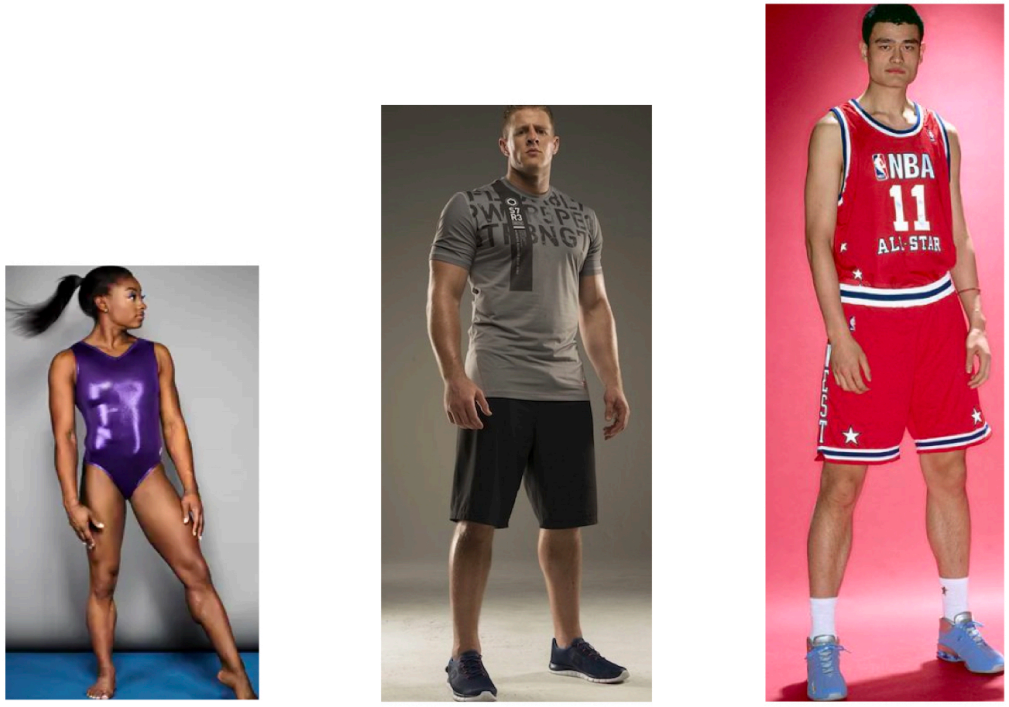


Figure 1: Three Houston sporting legends with their raw heights: Simone Biles 56 inches, JJ Watt 77 inches, and Yao Ming 90 inches.

Baseline/offset form

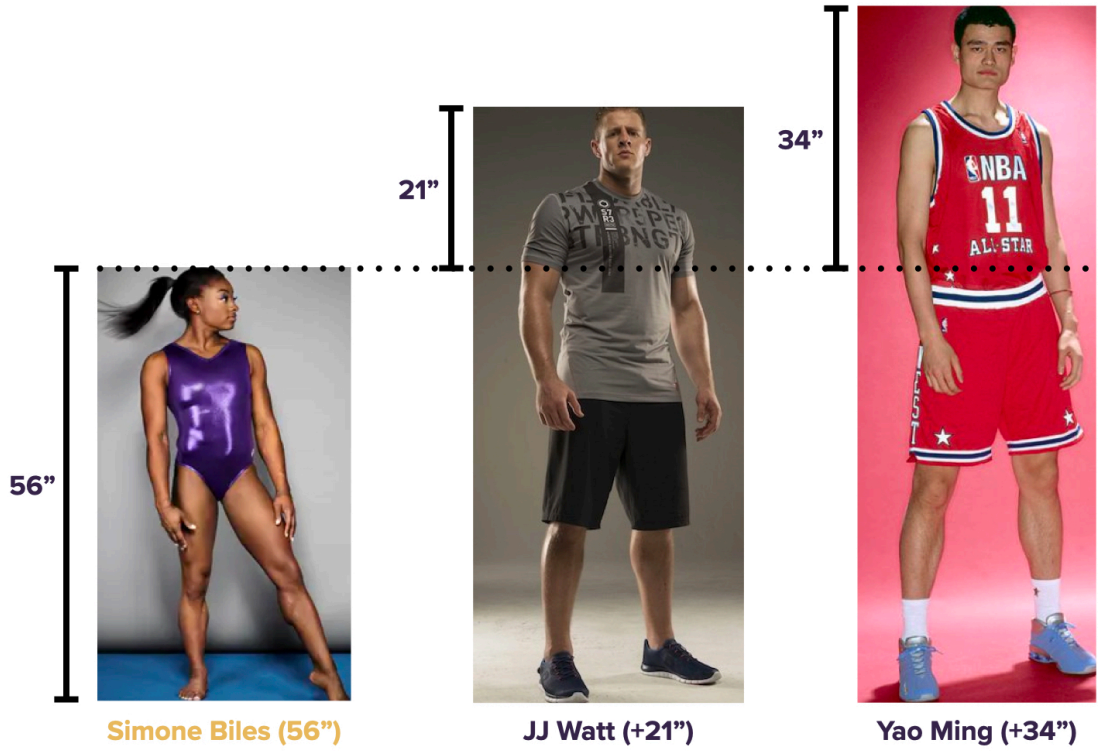


Figure 2: The same three athletes expressed with Simone Biles as the baseline: Simone Biles 56 inches, JJ Watt plus 21 inches, Yao Ming plus 34 inches.

A different baseline

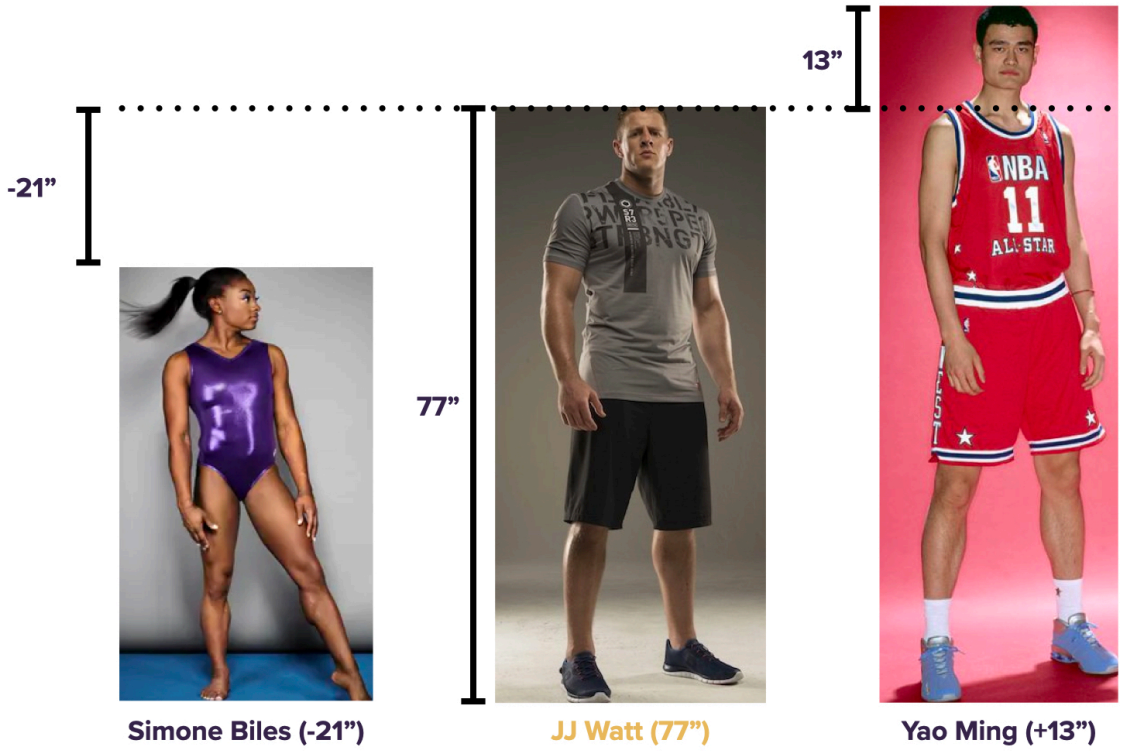


Figure 3: The same three athletes expressed with JJ Watt as the baseline: Simone Biles minus 21 inches, JJ Watt 77 inches, Yao Ming plus 13 inches.

Baseline/offset form

Raw form:

- Simone Biles: 56 inches
- JJ Watt: 77 inches
- Yao Ming: 90 inches

Baseline/offset form, using Simone Biles as the baseline:

- Simone Biles: 56 inches
- JJ Watt: +21 inches
- Yao Ming: +34 inches

Same information. Different encoding.

Why do this?

Baseline/offset form is awkward for three numbers.

It becomes useful when the model has:

- many groups,
- multiple categorical variables,
- numerical variables and categorical variables together,
- interactions.

Most regression software calls the baseline the intercept, even if geometrically, there's no line.

Cheese at Kroger

The data: 61 weeks of packaged cheese sales at a Dallas-area Kroger.

The question:

How much more cheese sells when there is an in-store ad display?

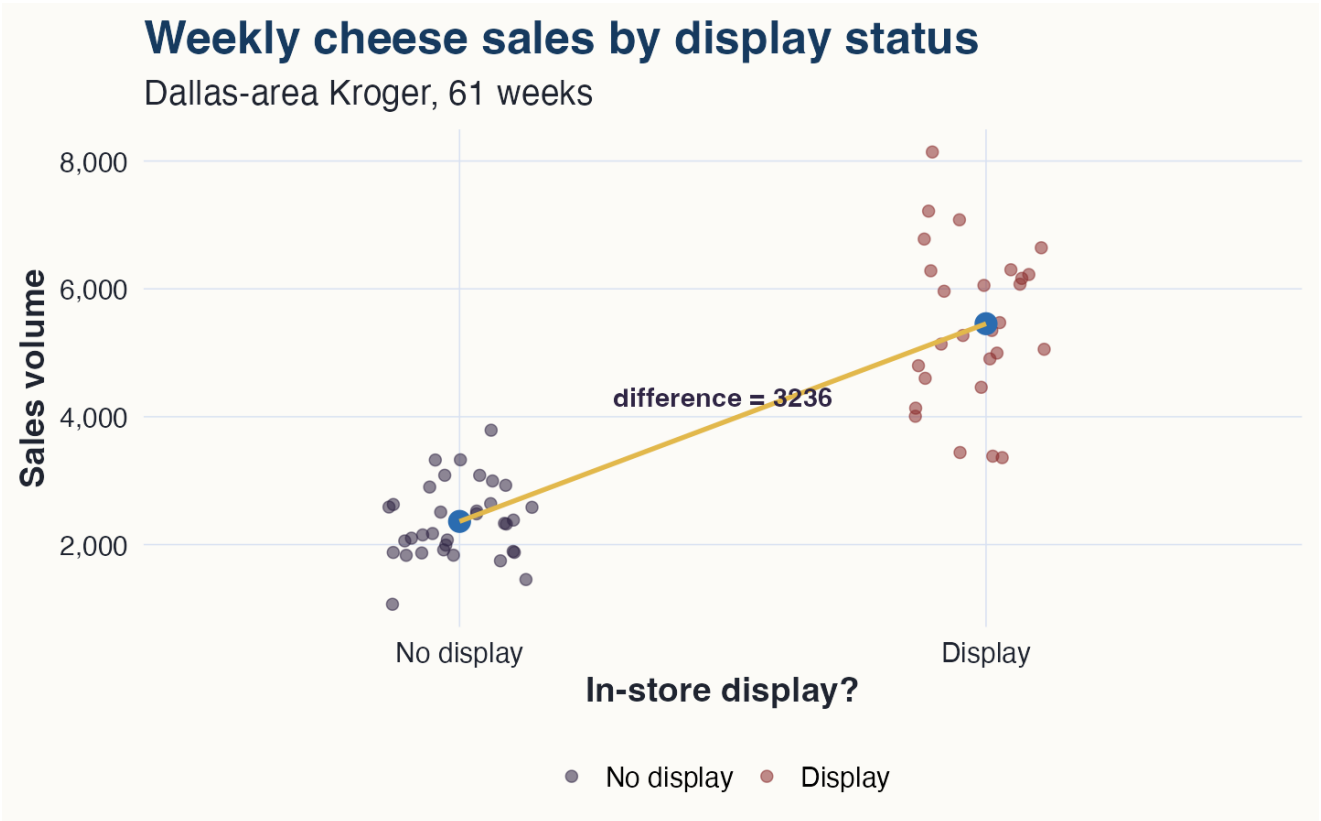


Figure 4: Jitter plot of weekly cheese sales by display status, with group

Cheese means

Average weekly sales:

display?	mean sales
no	2341
yes	5577

Same information in baseline/offset form:

group	baseline/offset
no display	2341
display	3236 higher

The display group is the baseline plus an offset.

Dummy variables

A dummy variable takes only two values:

- 1 if a case is in some category,
- 0 otherwise.

For cheese:

$$\text{Display}_i = \begin{cases} 1, & \text{week } i \text{ had a display} \\ 0, & \text{otherwise} \end{cases}$$

The dummy turns a yes/no question into a (binary) number.

The cheese model

Use the dummy variable as a predictor:

$$\text{Sales}_i = \beta_0 + \beta_1 \text{Display}_i + e_i.$$

If Display $= 0$:

$$\text{Sales}_i = \beta_0 + e_i.$$

If Display $= 1$:

$$\text{Sales}_i = \beta_0 + \beta_1 + e_i.$$

Reading the coefficients

For the cheese data:

$$\widehat{\text{Sales}} = 2341 + 3236 \cdot \text{Display}.$$

Interpretation:

- intercept: predicted mean sales without a display,
- display coefficient: predicted difference between display and no-display weeks,
- fitted display mean: $2341 + 3236 = 5577$.

The coefficient is the comparison we wanted.

Let your software create the dummies

We can create the dummy by hand:

```
mutate(dispyes = ifelse(disp == "yes", 1, 0))
lm(vol ~ dispyes, data = kroger_dallas)
```

But your software will do it automatically:

```
lm(vol ~ disp, data = kroger_dallas)
```

The software chooses a baseline level and creates the offset.

More than two groups

Suppose a categorical variable has K groups.

Regression uses:

- one baseline group (e.g. Simone Biles)
- $K-1$ dummy variables (e.g. one each for JJ Watt and Yao Ming),
- $K-1$ offsets from the baseline.

Why not K dummies?

- Because one group must be the reference point.
- Otherwise the model encodes the same information twice.

The baseline is arbitrary

Changing the baseline changes the coefficients.

It does not change the fitted group means.

If we use a different baseline:

- the intercept changes,
- the offsets change,
- the predictions stay the same.

So choose a baseline that makes the model easy to explain.

More than one categorical variable

A delivery platform is testing priority dispatch. Each row is an order.

```
delivery %>% sample_n(5)
```

order	rest	Peak	Priority	minutes late
1800	R06	0	0	5.7
685	R08	1	1	7.8
1536	R08	0	0	3.5
558	R11	0	0	5.5
1303	R02	1	1	14.6

Business question:

How much does Priority dispatch reduce late deliveries?

Start simple

Let's start by asking the simplest possible question: **Are priority orders earlier, on average?**

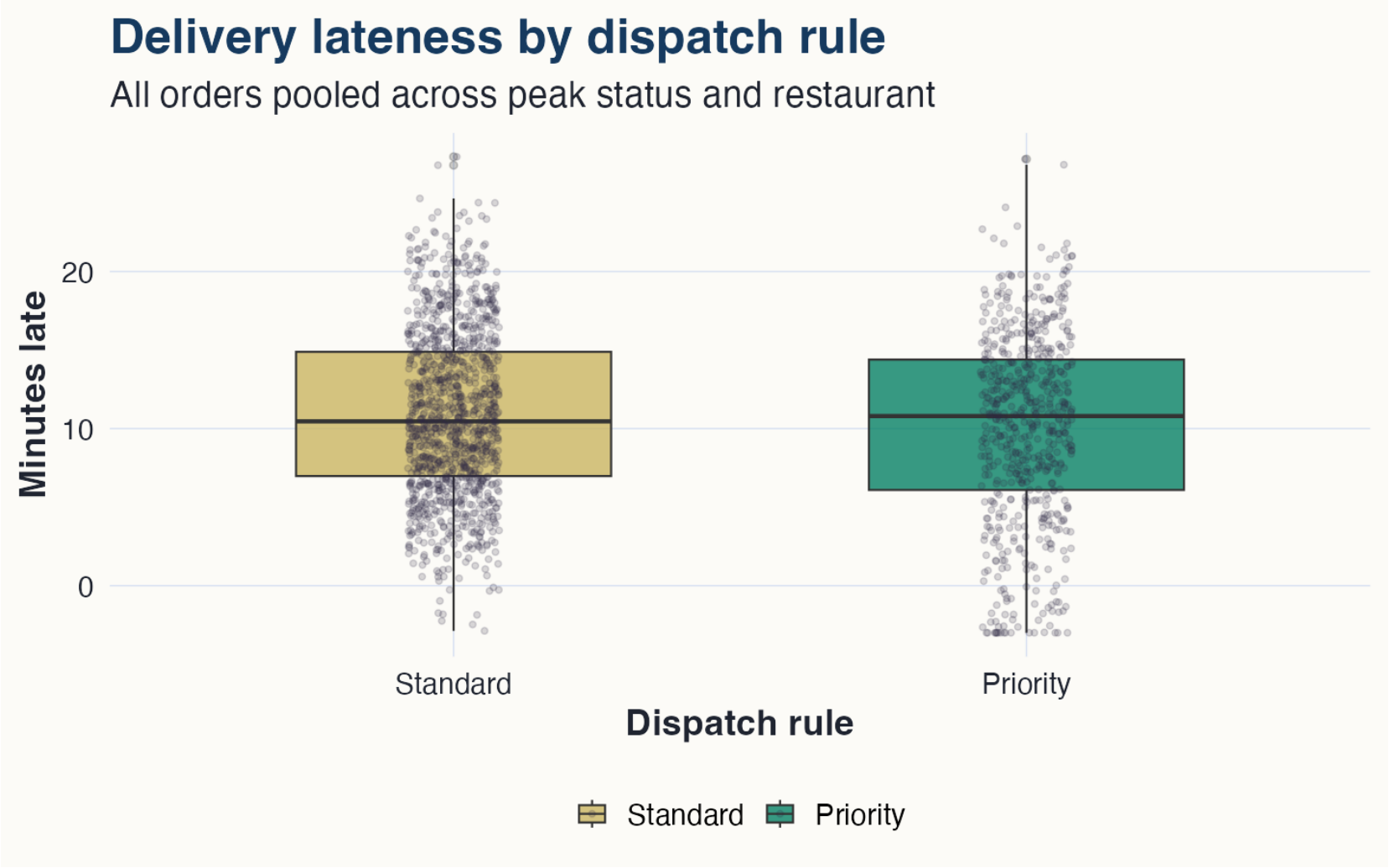


Figure 5: Boxplots of delivery lateness by standard versus priority dispatch

The one-variable model

Fit the model we already know how to fit:

$$\text{Late}_i = \beta_0 + \beta_1 \text{Priority}_i + e_i.$$

For these data:

$$\widehat{\text{Late}} = 11.2 - 0.9\text{Priority}.$$

That says priority dispatch predicts only about 0.9 fewer minutes late.

The one-variable model

Fit the model we already know how to fit:

$$\text{Late}_i = \beta_0 + \beta_1 \text{Priority}_i + e_i.$$

For these data:

$$\widehat{\text{Late}} = 11.2 - 0.9\text{Priority}.$$

That says priority dispatch predicts only about 0.9 fewer minutes late.

Pretty underwhelming. Is priority dispatch just a fancy button likely to disappoint customers in the long run?

But look at when priority dispatch is used

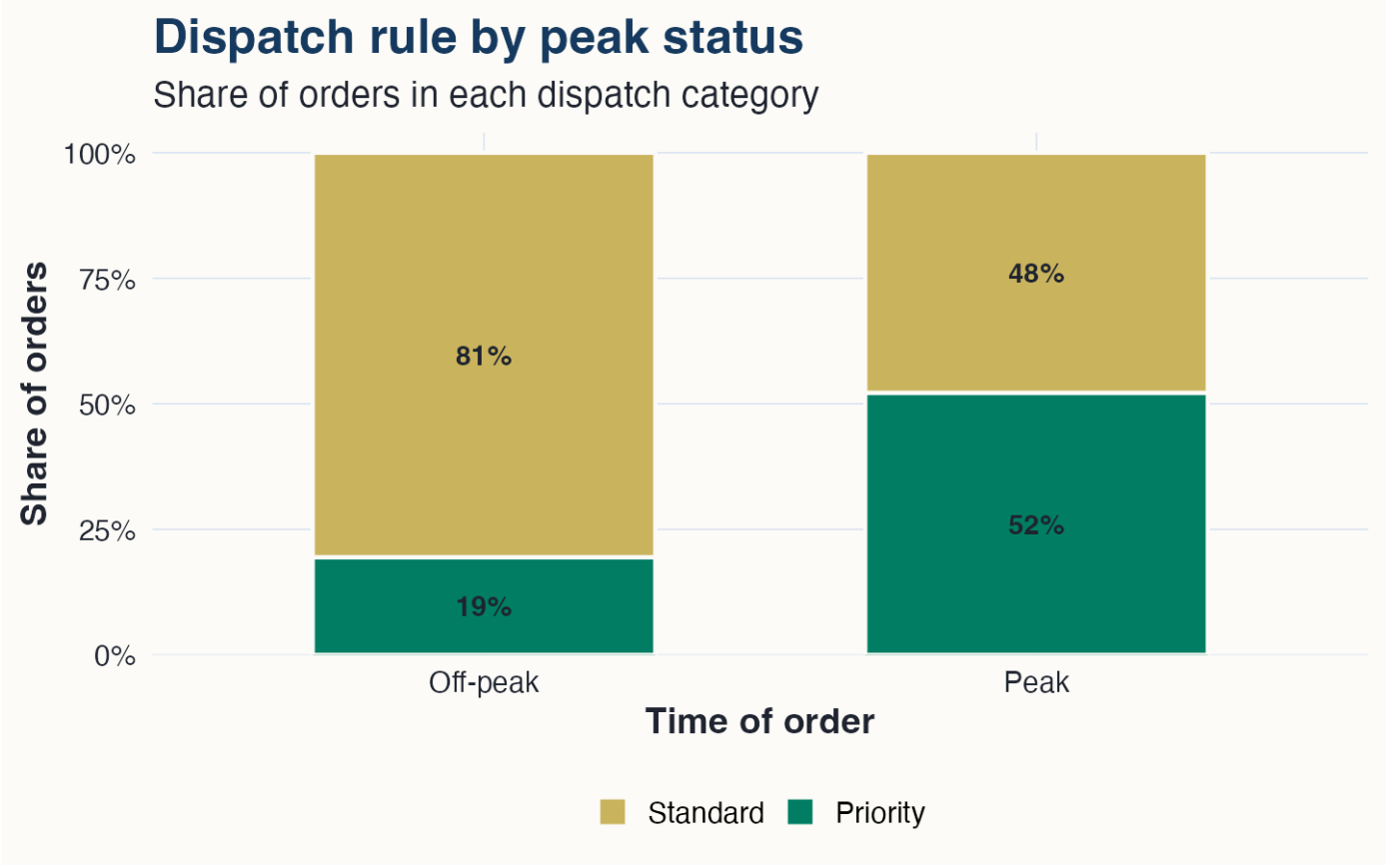


Figure 6: Priority dispatch is used far less frequently for off-peak orders.

But look at when priority dispatch is used

So our naive comparison is not just **priority versus standard.**

It is also, implicitly, **peak-hour versus off-peak orders.**

But look at when priority dispatch is used

So our naive comparison is not just **priority versus standard.**

It is also, implicitly, **peak-hour versus off-peak orders.**

This is **confounding.** A confounder is a variable that is related to:

- the predictor we care about,
- the outcome we are trying to explain.

So the one-variable priority coefficient mixes two stories.

Adjust for peak hours

Use both dummies:

$$\text{Late}_i = \beta_0 + \beta_1 \text{Priority}_i + \beta_2 \text{Peak}_i + e_i.$$

Side note: this is the moment regression becomes a language for comparisons with multiple inputs:

$$\widehat{y} = f(x_1, x_2, ...).$$

We'll lean on this heavily from now on!

The adjusted fit

For these data:

$$\widehat{\text{Late}} = 8.9 - 3.8\text{Priority} + 7.6\text{Peak}.$$

Now the priority coefficient looks more sensible:

- compare two off-peak orders, or
- compare two peak-hour orders,
- holding peak status fixed.

Priority dispatch predicts about 3.8 fewer minutes late.

Adjustment as subtraction

Start with the fitted equation:

$$\widehat{\text{Late}} = 8.9 - 3.8\text{Priority} + 7.6\text{Peak}.$$

Move the peak-hour piece to the left:

$$\widehat{\text{Late}} - 7.6\text{Peak} = 8.9 - 3.8\text{Priority}.$$

Left side: delivery lateness, adjusted for peak status.

Right side: the partial priority effect.

Partial versus overall

Overall relationship:

Compare all priority orders with all standard orders.

That gave about -0.9 minutes.

Partial relationship:

Compare priority and standard orders with the same peak status.

That gave about -3.8 minutes. Different comparison, different coefficient.

Main effects only

Our two-variable model was:

$$\text{Late} \sim \text{Priority} + \text{Peak}.$$

This is a main-effects model.

It assumes the priority gap is the same regardless of circumstance:

- off-peak,
- at peak.

Does the gap look the same?

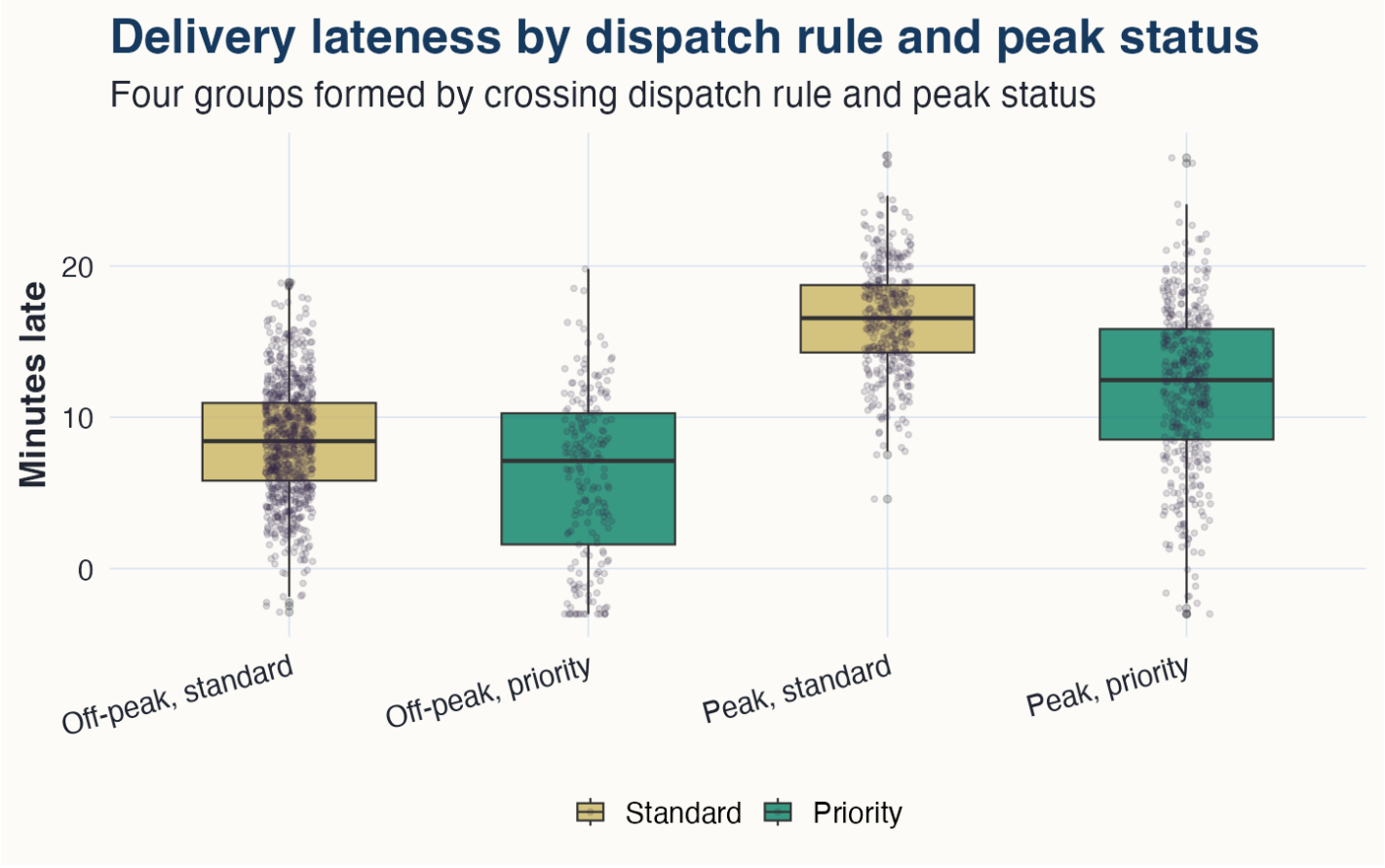


Figure 7: Boxplots of delivery lateness for four groups: off-peak standard, off-peak priority, peak standard, and peak priority. Priority dispatch helps more at peak.

What is an interaction?

An interaction means the effect of one variable depends on another variable.

Plain examples:

- high gear is fine on flat ground, awful uphill,
- a breeze feels good in August, bad in January,
- two Tylenol pills mean one thing for an adult and another thing for an elephant.

The phrase "the effect of x on y" is incomplete until we name the context in which x operates.

Add the interaction

We encode interactions using the product of the two dummy variables:

$$\text{Late}_i = \beta_0 + \beta_1 \text{Priority}_i + \beta_2 \text{Peak}_i + \beta_{12} \text{Priority}_i \text{Peak}_i + e_i.$$

The product is 1 only for priority orders during peak hours.

That last coefficient is the extra "oomph" from being both Priority=1 and Peak=1.

Let's see why this works.

Four fitted means

With the interaction:

Priority	Peak	fitted mean
0	0	β_0
1	0	$\beta_0 + \beta_1$
0	1	$\beta_0 + \beta_2$
1	1	$\beta_0 + \beta_1 + \beta_2 + \beta_{12}$

The priority effect is now context-specific:

- off-peak: β_1,
- peak: $\beta_1 + \beta_{12}$.

Interaction fit

For these data:

$$\widehat{\text{Late}} = 8.7 - 2.6\text{Priority} + 8.3\text{Peak} - 2.1\text{Priority} \cdot \text{Peak}.$$

Then:

- priority effect off-peak: -2.6 minutes,
- priority effect at peak: $-2.6 - 2.1 = -4.7$ minutes.

So what's the effect of Priority dispatch? **It depends.** That is the interaction.

But restaurants differ, too

Some restaurants are just harder:

- slower kitchen,
- parking chaos,
- awkward pickup counter,
- local traffic patterns.

"Which restaurant" is not the main business question. But ignoring it can still distort the answer.

But restaurants differ, too

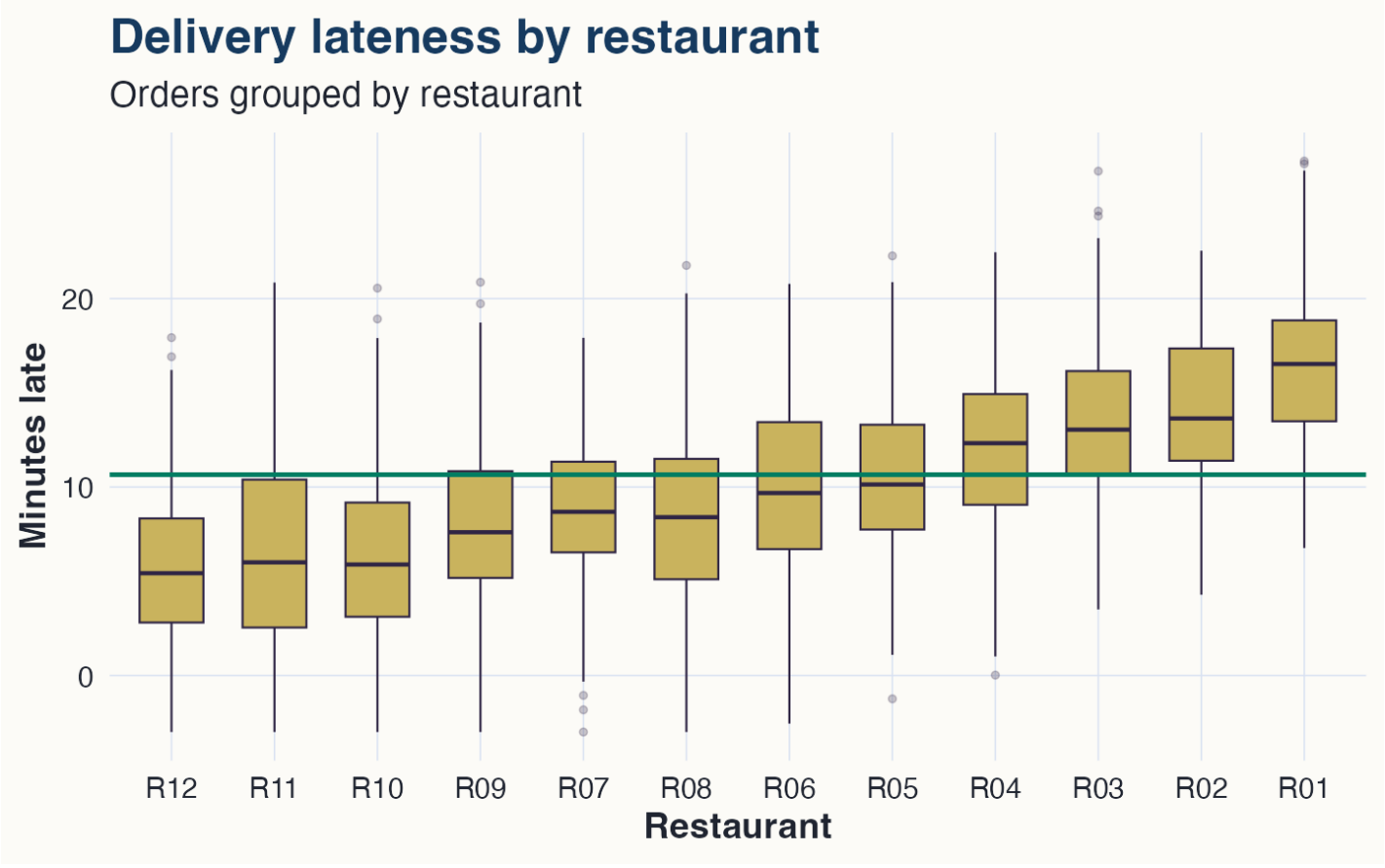


Figure 8: Boxplots of delivery lateness across 12 restaurants. Some restaurants are consistently later than others.

Restaurant-level confounding

Priority dispatch also differs by restaurant.

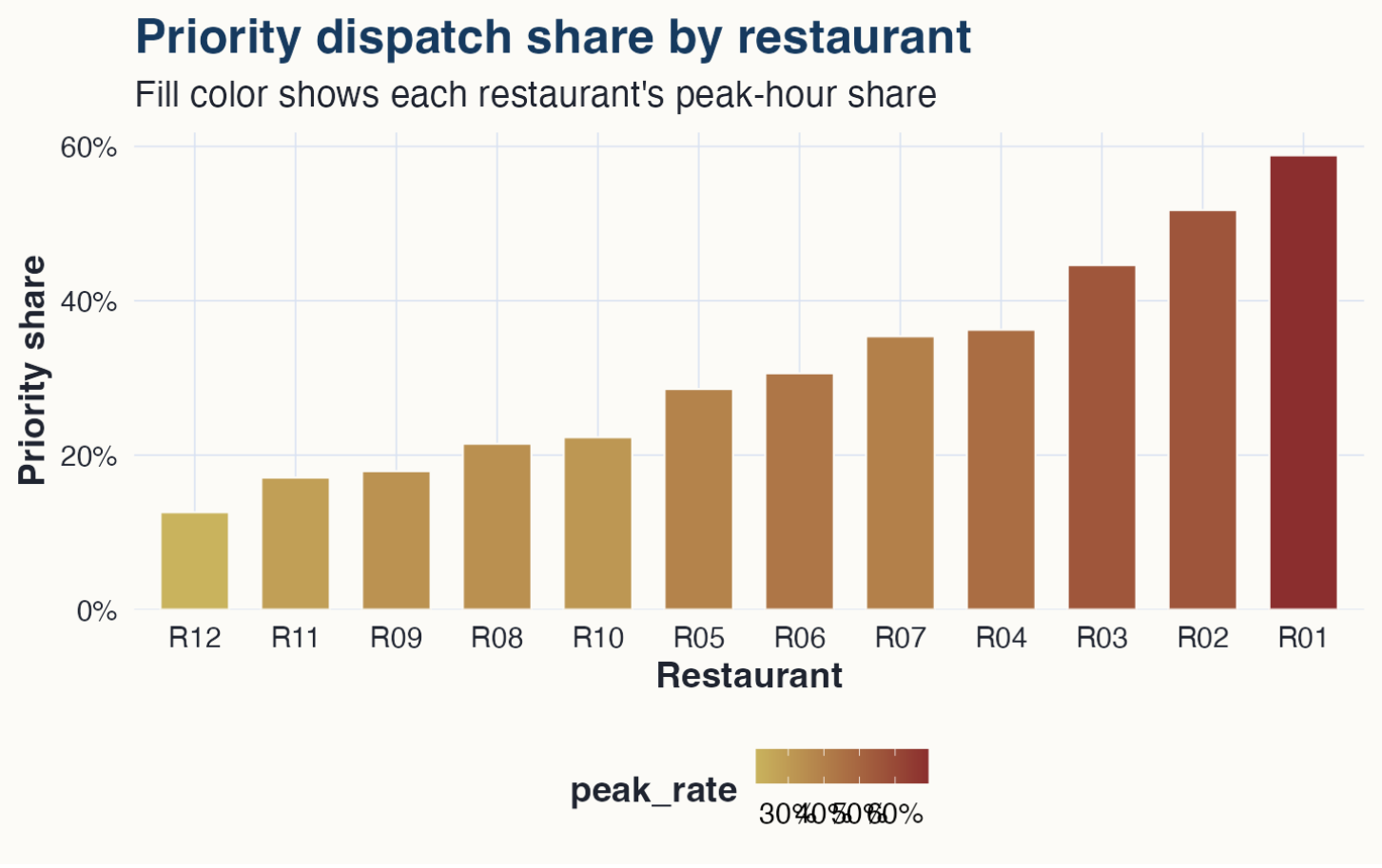


Figure 9: Bar chart of restaurant priority dispatch rates, with fill showing peak-hour share.

Restaurant-level confounding

So the priority comparison is implicitly mixing over:

- dispatch rule,
- peak-hour mix,
- restaurant mix.

This is why observational data are a pain in the neck.

Add restaurant dummies

Use restaurant offsets:

$$\text{Late} \sim \text{Priority} + \text{Peak} + \text{Priority} : \text{Peak} + \text{Restaurant}.$$

Now the restaurant coefficients adjust baseline lateness.

That is a baseline shift, not a new dispatch effect.

Regression output

Fitted model:

$$\text{Late} \sim \text{Priority} + \text{Peak} + \text{Priority} : \text{Peak} + \text{Restaurant}.$$

coefficient	estimate
intercept: R01, standard, off-peak	14.3
priority offset	-4.0
peak offset	7.5
priority:peak offset	-2.1

These are the coefficients tied directly to dispatch and peak status.

Restaurant offsets

Same model, restaurant coefficients:

restaurant	offset	restaurant	offset
R02	-1.7	R08	-7.8
R03	-2.9	R09	-7.5
R04	-3.3	R10	-8.5
R05	-4.6	R11	-9.1
R06	-6.1	R12	-9.3
R07	-6.3		

All restaurant offsets are relative to R01. For example, R12 is about 9.3 minutes earlier than R01, after adjusting for dispatch and peak status.

The equation gets ugly

With 12 restaurants, the full model equation looks like:

$$\begin{aligned}\text{Late}_i = \beta_0 + \beta_1 \text{Priority}_i + \beta_2 \text{Peak}_i + \beta_{12} \text{Priority}_i \text{Peak}_i \\ + \gamma_2 R02_i + \gamma_3 R03_i + \cdots + \gamma_{12} R12_i + e_i.\end{aligned}$$

(Self-check: why only 11 restaurant dummies?)

This is why nobody enjoys reading raw regression output before coffee.

Back to baseline/offset form

This is where baseline/offset form starts earning its keep.

Question:

How does priority dispatch affect delivery delay?

There are two paths:

- treat every group mean as primary,
- treat a baseline and offsets as primary.

The second path is how regression stays readable as the problem gets more complicated.

Path 1: raw means

Raw form says: **compute the mean lateness for every combination.**

Here that means:

12 restaurants $\times 2$ peak statuses $\times 2$ dispatch rules $= 48$ cell means.

Then we stare across those 48 means and try to answer: **what is the priority effect?**

That is not a clean question anymore.

What goes wrong

The raw-means path is clumsy.

- Some cells might have very small samples. (In these data, the smallest cell has only 5 orders.)
- The priority effect has to be reconstructed by comparing many pairs of means.
- Adding another variable means slicing again.

It also implicitly allows every effect to depend on everything else:

$$\text{Priority} \times \text{Peak} \times \text{Restaurant}.$$

That is a three-way interaction.

Path 2: baseline plus offsets

Baseline/offset form says: **build each fitted mean out of components.**

For example:

$$\begin{aligned}\widehat{\text{Late}} = \ & \text{baseline lateness} \\ & + \text{priority offset} \\ & + \text{peak offset} \\ & + \text{restaurant offset} \\ & + \text{chosen interaction offsets.}\end{aligned}$$

The fitted means are still there implicitly; they are just assembled from *comparisons among conditions*.

Why this helps

Regression software does the assembly. We get a direct answer to the question/contrast we care about:

> *priority versus standard dispatch, holding peak status and restaurant fixed.*

That is much cleaner once the problem has several moving parts.

One more interaction

What if priority dispatch works better for some restaurants than others?

Maybe:

- one restaurant has a dedicated pickup shelf,
- another makes drivers wait in line,
- another is already so fast that priority has little room to help.

Then the priority effect depends on restaurant.

Add `Priority:Restaurant`.

Full working model

The model formula:

Late $\sim$ Priority+Peak+Restaurant+Priority : Peak+Priority : Restaurant.

This includes two two-way interactions:

- `Priority:Peak`,
- `Priority:Restaurant`.

It lets the priority effect depend on peak status and restaurant.

Reading the priority coefficients

From the fitted model:

coefficient	estimate
priority effect at R01, off-peak	-0.7
extra priority effect at peak	-2.8
extra priority effect at R09	-6.4
extra priority effect at R10	-6.8
extra priority effect at R11	-6.7
extra priority effect at R12	-6.3

For any restaurant:

$$\text{priority effect at peak} = \text{priority effect off-peak} - 2.8.$$

Negative means fewer minutes late.

Where priority saves the most time

The model-implied priority savings:

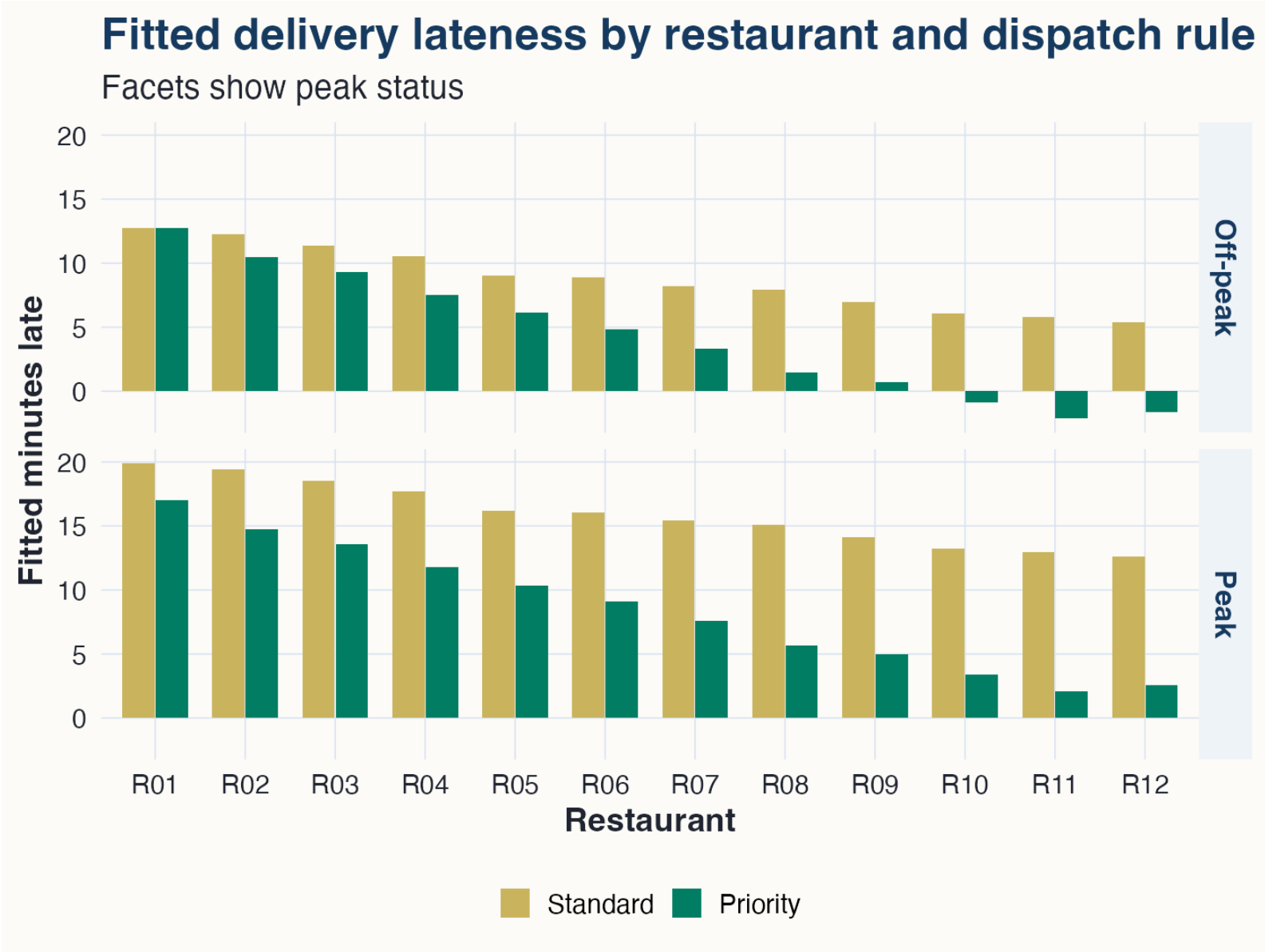
restaurant	off-peak savings	peak savings
R10	7.5 min	10.3 min
R11	7.4 min	10.2 min
R09	7.1 min	9.9 min
R12	7.0 min	9.8 min
R07	5.9 min	8.7 min

These are fitted contrasts:

standard dispatch minus priority dispatch, holding restaurant and peak status fixed.

Visualize the fitted effects

Instead of reading 20+ coefficients, plot the fitted means.



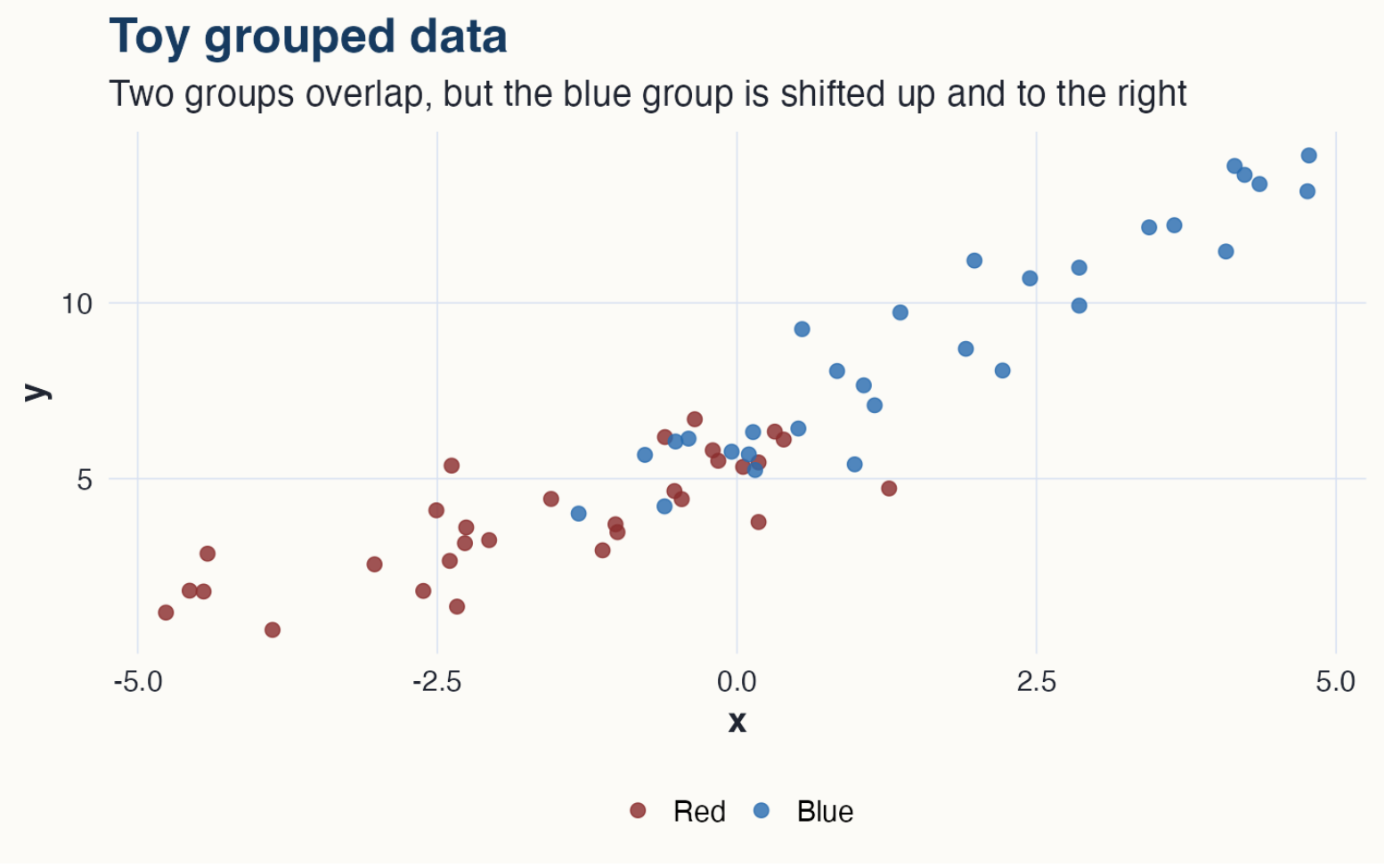
Numerical and categorical variables together

Now we mix two kinds of predictors:

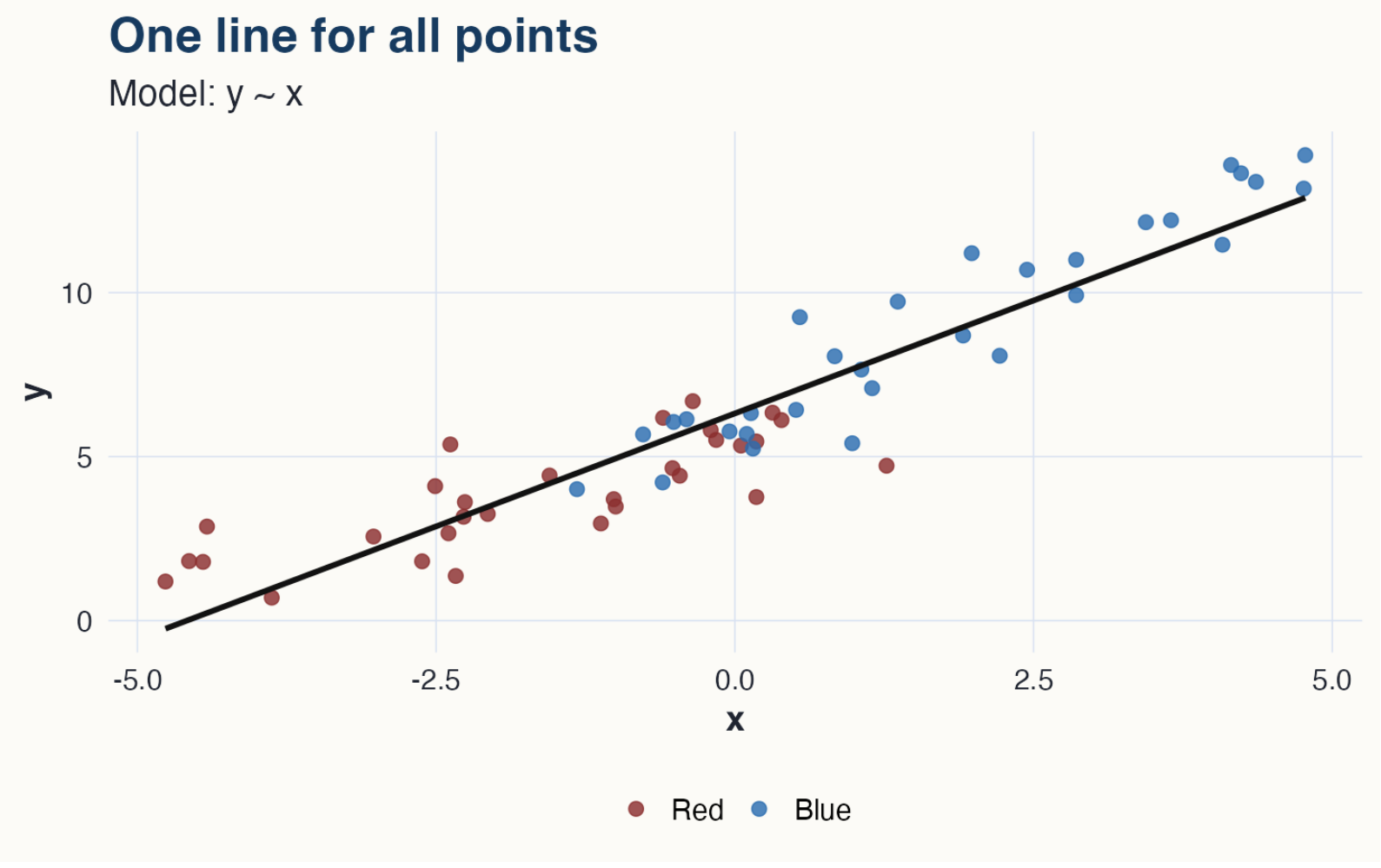
- numerical variables: slopes,
- categorical variables: baseline shifts,
- interactions: context-specific slopes.

Start with a toy picture.

Toy grouped data



One line ignores the groups



The one-line model

The model:

$$y_i = \beta_0 + \beta_1 x_i + e_i.$$

This treats the red and blue points as one undifferentiated cloud: same intercept, same slope.

But the blue points look shifted up, relative to the red, even at the same value of x.

Add the dummy

A model with a dummy variable captures this:

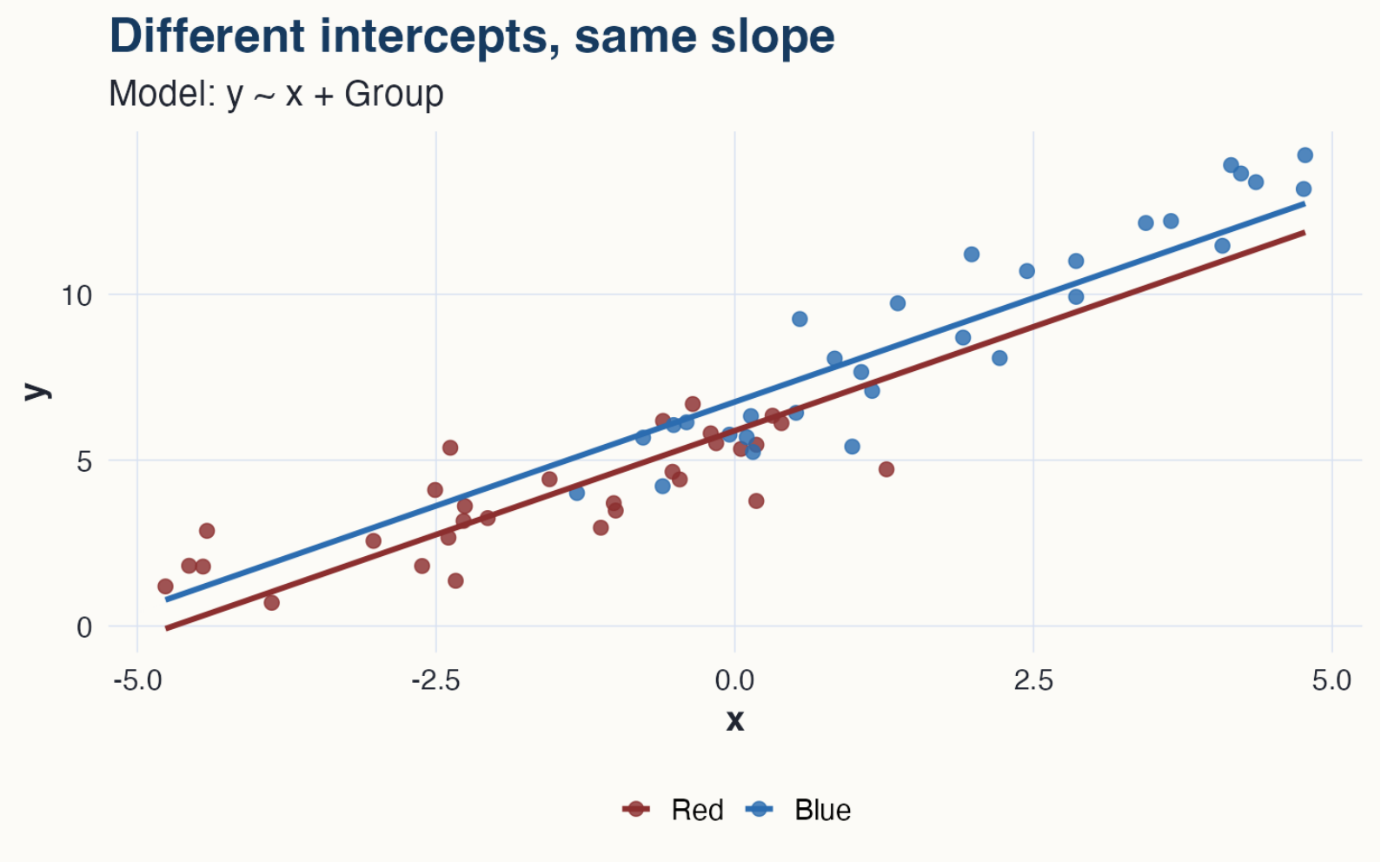
$$y_i = \beta_0 + \beta_1 x_i + \beta_2 \text{Blue}_i + e_i.$$

Now blue gets an intercept offset.

- Red equation: $\hat{y} = \beta_0 + \beta_1 x$
- Blue equation: $\hat{y} = (\beta_0 + \beta_2) + \beta_1 x$

The $\beta_2 \text{Blue}_i$ term shifts the line up whenever $\text{Blue}_i = 1$.

Add the dummy



What is missing?

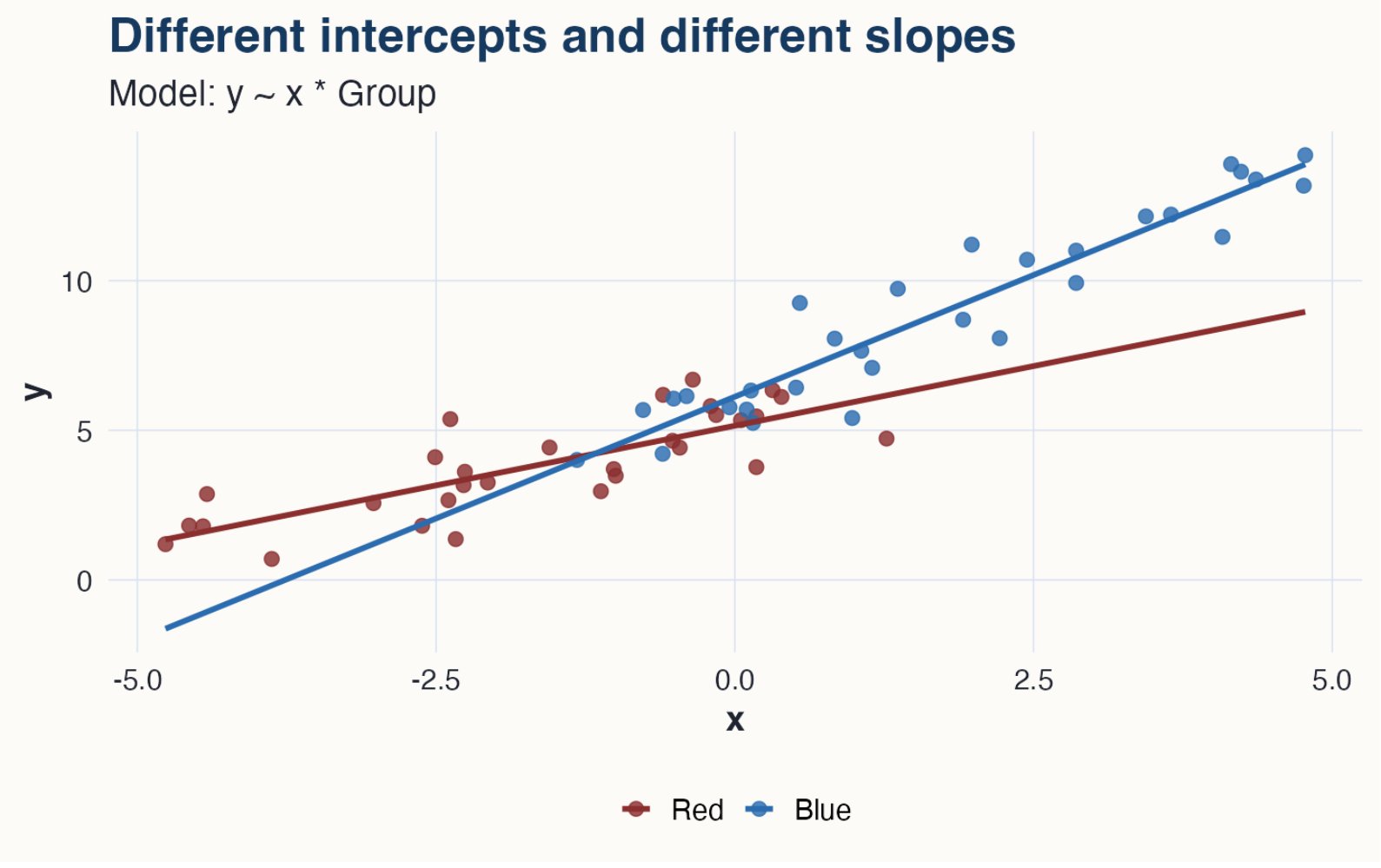
But the two fitted lines are still parallel.

The picture suggests two things:

- blue starts higher,
- blue rises faster with x.

A dummy variable handles the first part, but not the second.

Add the interaction



The interaction model

The model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 \text{Blue}_i + \beta_3 x_i \text{Blue}_i + e_i.$$

For red points, Blue $= 0$:

$$\hat{y} = \beta_0 + \beta_1 x.$$

For blue points, Blue $= 1$, we collect like terms:

$$\hat{y} = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x.$$

The dummy coefficient is an intercept offset; the interaction coefficient is a slope offset.

Austin house prices

A sample of Austin home sales, collected by students for a class project last year, focusing on three neighborhoods, post 2023:

- Hyde Park,
- Clarksville,
- Barton Hills.

Variables:

- `Sqft`: square footage,
- `PriceK`: sale price in thousands,
- `MajorReno10`: major renovation in the last 10 years.

Price per square foot

Realtors often quote:

$$\frac{\text{price}}{\text{square footage}}.$$

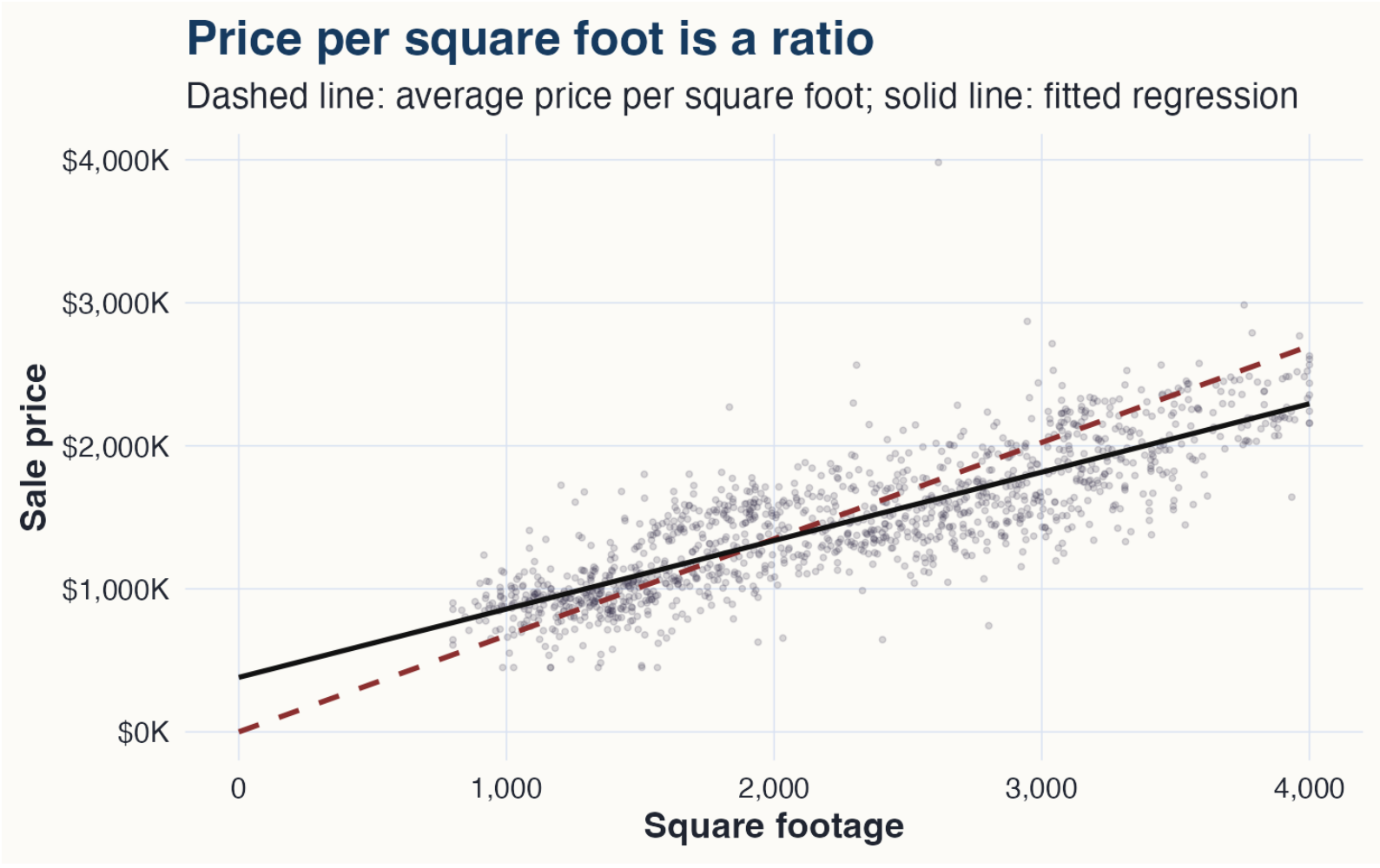
That sounds like a slope: dollars per extra square foot.

But it is a **ratio**, not a marginal slope.

It draws a line through zero:

$$\text{Price} = (\text{price per sqft}) \cdot \text{Sqft}.$$

Ratio versus fitted line



Diagnosing confounders

But fitting a single line through the point cloud potentially mixes over multiple stories:

- size
- neighborhood
- renovation status

Let's check whether we should worry about confounders here.

EDA: neighborhood

Size and price differ by neighborhood:

neighborhood	mean sqft	mean price	mean price/sqft
Hyde Park	2065	1194K	$604
Clarksville	2000	1289K	$689
Barton Hills	2676	1870K	$727

Barton Hills has larger houses and higher prices.

Clarksville has similar average size to Hyde Park, but higher average price.

Neighborhood cannot be ignored.

EDA: renovation status

Size and price also differ by renovation status:

major renovation	mean sqft	mean price	mean price/sqft
No	1979	1297K	$690
Yes	2692	1717K	$650

Recently renovated homes are larger, and also more expensive, on average.

So renovation status is also confounded with size.

Building a model

So these three variables are all confounded with each other:

- size
- neighborhood
- renovation status

Let's build a model that cleanly separates these effects. Remember:

- dummy variables shift the line up or down
- interactions between dummies and numerical variables change the slope

Centering the predictor

We need a preliminary data-wrangling step. Define:

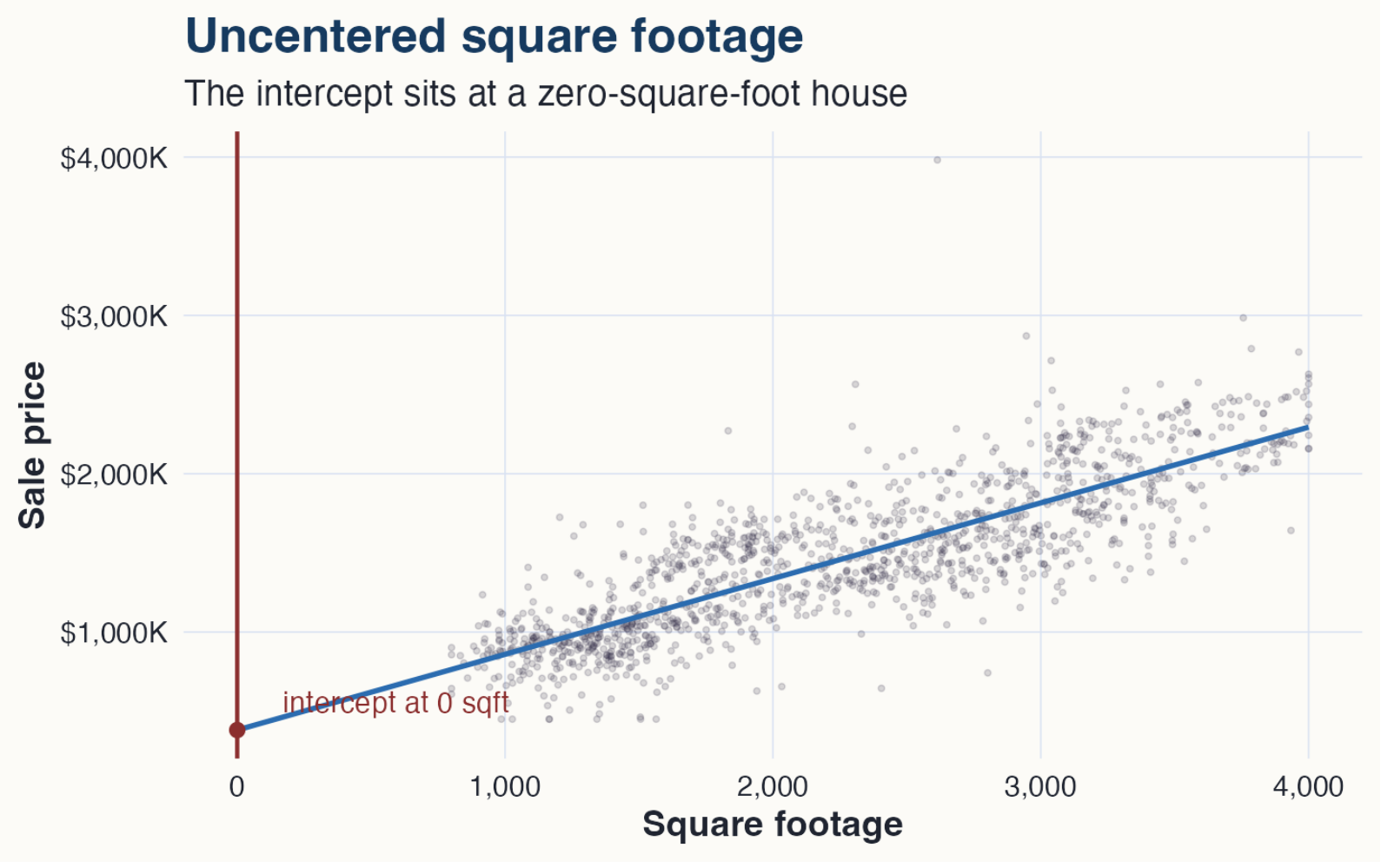
$$\text{Sqft100} = \frac{\text{Sqft} - 1800}{100}.$$

Why subtract 1800? Because 1800 sq.ft. is a typical house in this sample. So:

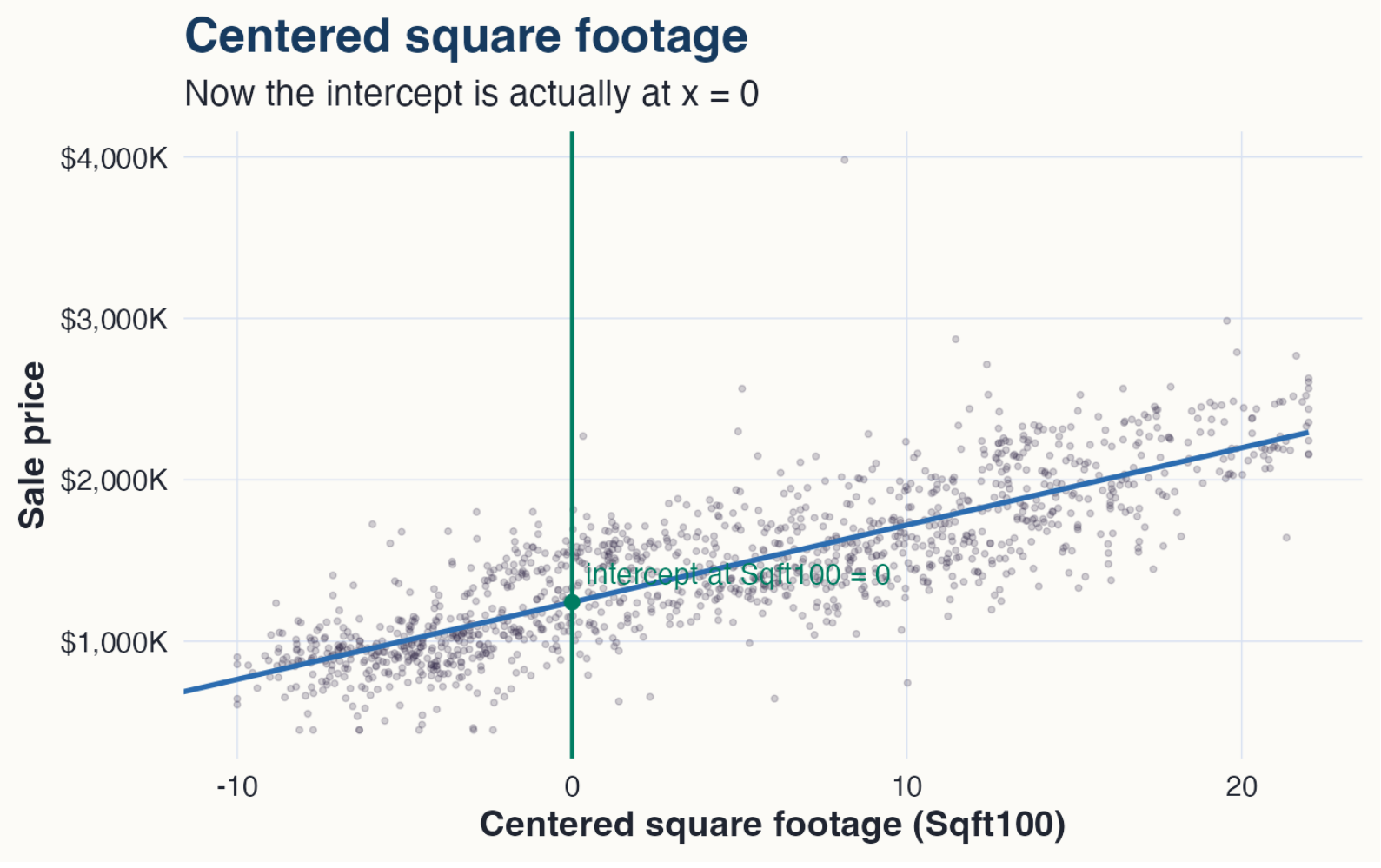
- `Sqft100 = 0` means 1800 square feet,
- `Sqft100 = 1` means 1900 square feet,
- `Sqft100 = -1` means 1700 square feet.

The slope is still measured per 100 square feet.

Without centering



With centering



Main effects model

Start with:

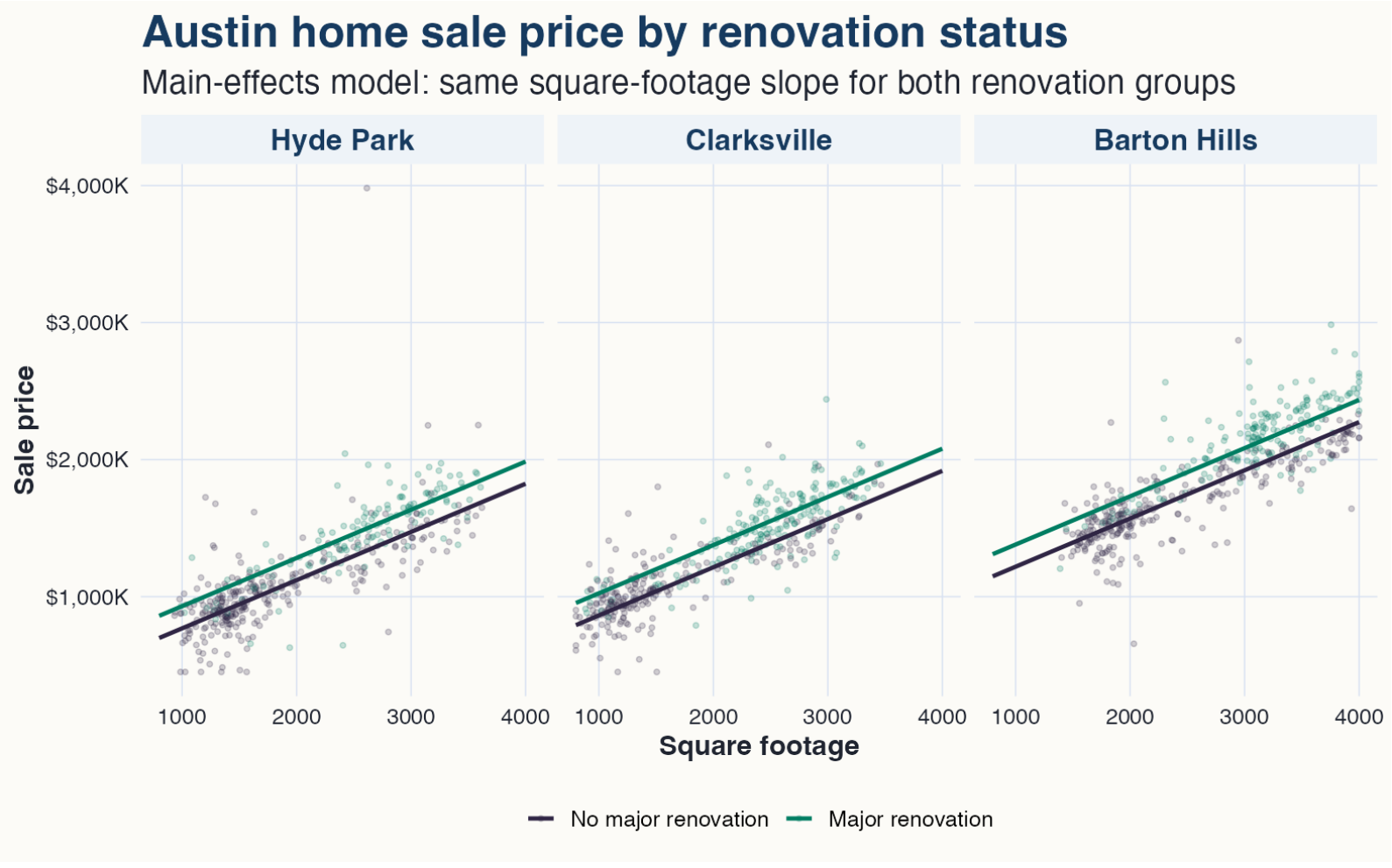
$$\text{PriceK} \sim \text{Sqft100} + \text{MajorReno10} + \text{Neighborhood}.$$

Our fitted model:

$\text{PriceK} = 1050 + 35.2\text{Sqft100} + 161.8\text{MajorReno10} + \text{neighborhood offsets} +$

This model shifts the line up or down by neighborhood and renovation status.

One model, six lines



Add the interaction

Now we let the square-footage slope differ by renovation status:

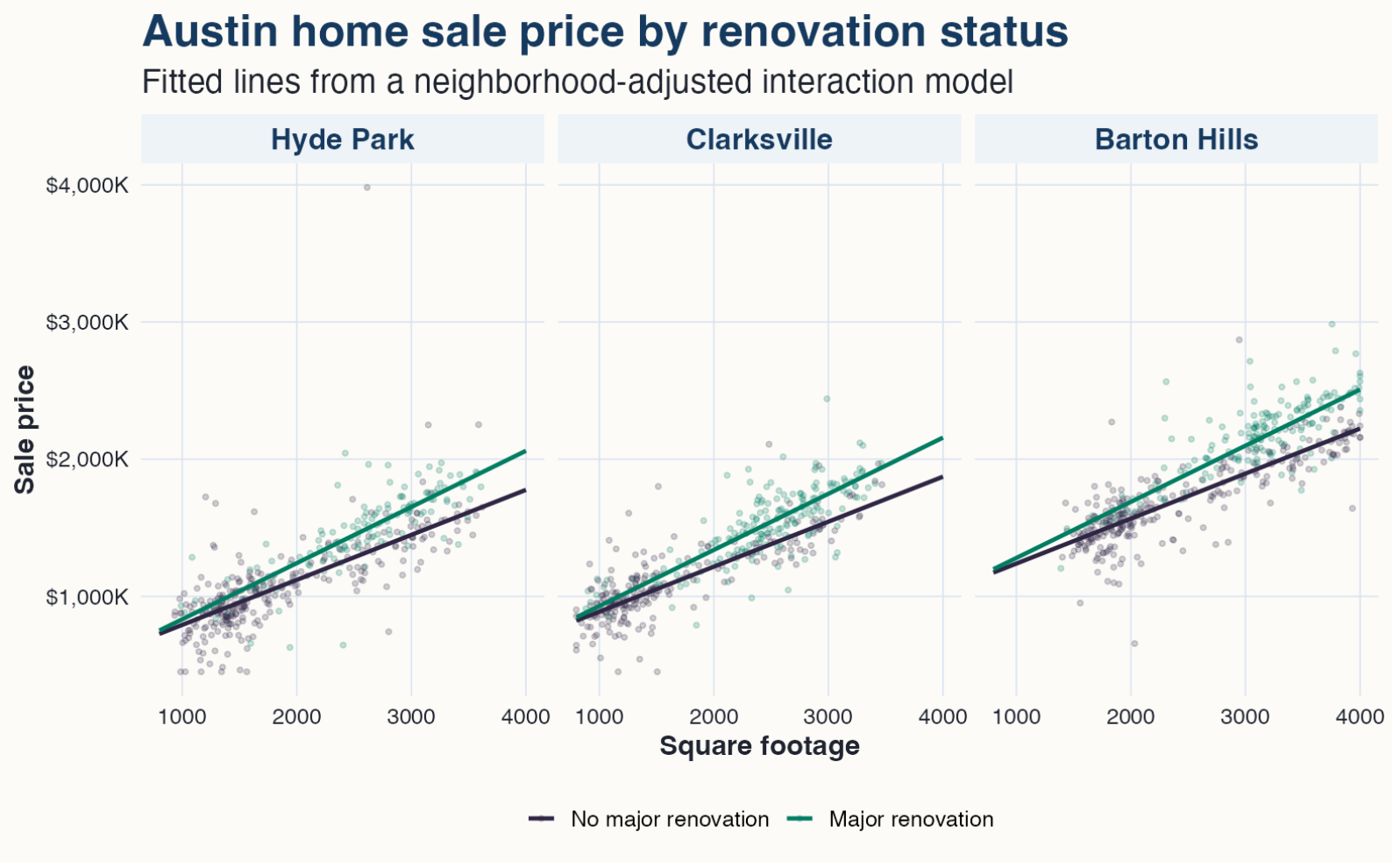
$$\text{PriceK} \sim \text{Sqft100} * \text{MajorReno10} + \text{Neighborhood}.$$

For these data:

$$\begin{aligned}\widehat{\text{PriceK}} = {} & 1054.6 + 105.9\text{MajorReno10} \\ & + 32.8\text{Sqft100} \\ & + 8.1\text{Sqft100} \cdot \text{MajorReno10} \\ & + \text{neighborhood offsets}\end{aligned}$$

Now renovation status shifts the intercept and the slope.

Separate slopes by renovation status



Hyde Park arithmetic

Let $R =$ MajorReno10, $C =$ Clarksville, and $B =$ Barton Hills. The full fitted equation is:

$$\widehat{\text{PriceK}} = 1054.6 + 32.8\text{Sqft100} + 105.9R + 96.1C + 446.7B + 8.1(\text{Sqft100} \cdot R).$$

Hyde Park: $C = 0$, $B = 0$. There are two lines:

- $R = 0$: $\widehat{\text{PriceK}} = 1054.6 + 32.8\text{Sqft100}$.
- $R = 1$: $\widehat{\text{PriceK}} = (1054.6 + 105.9) + (32.8 + 8.1)\text{Sqft100}$.

Barton Hills arithmetic

Let $R =$ MajorReno10, $C =$ Clarksville, and $B =$ Barton Hills. The full fitted equation is:

$$\widehat{\text{PriceK}} = 1054.6 + 32.8\text{Sqft100} + 105.9R + 96.1C + 446.7B + 8.1(\text{Sqft100} \cdot R).$$

Barton Hills: $C = 0$, $B = 1$. There are two lines:

- $R = 0$: $\widehat{\text{PriceK}} = (1054.6 + 446.7) + 32.8\text{Sqft100}$.
- $R = 1$: $\widehat{\text{PriceK}} = (1054.6 + 446.7 + 105.9) + (32.8 + 8.1)\text{Sqft100}$.

What does 105.9 mean?

The coefficient on `MajorReno10` is 105.9. The tempting reading is:

recent major renovation is worth \$105.9K.

But that is not quite right. With the interaction, the renovation gap is a difference between two fitted prices for houses of the same size, in the same neighborhood.

Fix a neighborhood and a value of `Sqft100`. Compare:

$$\begin{aligned}\widehat{\text{PriceK}}_{\text{reno}} &= \text{same neighborhood offset} + 1054.6 + 32.8\text{Sqft100} \\ &\quad + 105.9 + 8.1\text{Sqft100}, \\ \widehat{\text{PriceK}}_{\text{no reno}} &= \text{same neighborhood offset} + 1054.6 + 32.8\text{Sqft100}.\end{aligned}$$

Subtract:

$$\begin{aligned}\text{renovation gap} &= \widehat{\text{PriceK}}_{\text{reno}} - \widehat{\text{PriceK}}_{\text{no reno}} \\ &= 105.9 + 8.1\text{Sqft100}.\end{aligned}$$

Renovation gap by size

Because `Sqft100 = 0` means 1800 square feet:

home size	Sqft100	fitted renovation gap
1400 sqft	-4	\$73.5K
1800 sqft	0	\$105.9K
2200 sqft	4	\$138.3K

So the dummy coefficient is not "the value of renovation" for every home.

It is the renovation gap at the centered size: 1800 square feet.

Summary

With a numerical variable and a group:

- dummy variable: intercept offset,
- interaction with the numerical variable: slope offset.

For Austin:

- neighborhood adjusts baseline prices,
- renovation status adjusts baseline prices,
- `Sqft100:MajorReno10` lets the square-footage premium differ by renovation status.

That is the same toy red/blue grammar, just with real variables.

Correlation versus interaction

Students often confuse these.

Correlation among predictors means: It is hard to tell which predictor is driving variation in y.

Interaction means: The relationship between one predictor and y depends on another predictor.

Correlation is causal confusion.

Interaction is causal complexity.

Tennis forehand setup

Suppose a coach is rebuilding your forehand. Two things can change:

- grip: **semi-western** versus **western**,
- stance: **side-on/neutral** versus **open**.

The outcome: how many times you hit the deep target area, out of 100 forehands.

A bad experiment

The coach gives you two blocks:

stance	semi-western grip	western grip
side-on/neutral	58	–
open	–	72

You did better with the second setup.

So what helped?

Complete correlation

In the bad experiment:

stance	semi-western grip	western grip
side-on/neutral	58	–
open	–	72

Grip and stance are perfectly correlated.

That is causal confusion from bad design.

A better experiment

Now the coach has you hit 100 forehands in all four conditions:

stance	semi-western grip	western grip
side-on/neutral	66	58
open	55	75

Now grip and stance are not correlated by design.

That is what lets us ask better questions.

What the full table says

The stance effect depends on the grip. For the semi-western grip:

$$55 - 66 = -11.$$

For the western grip:

$$75 - 58 = 17.$$

Same stance change, different effect.

That is an interaction.

Neither implies the other

You can have:

- correlated predictors with no obvious interaction,
- an interaction in a balanced experiment,
- correlated predictors and an interaction.

So when someone asks, "what is the effect of x on y?"

Correlation means: hard to tell.

Interaction says: it depends.

Diagnosing interactions

Good reasons to include an interaction:

- subject-area knowledge says an effect is context-specific,
- a plot shows clearly different slopes or gaps,
- the interaction coefficient is practically meaningful,
- predictive performance improves enough to matter.

There is no universal cutoff.

You have to judge the problem.

The practical lesson

Categorical variables are not a special regression species.

They are a coding system.

- Dummies create baseline shifts.
- Multiple dummies create multiple offsets.
- Numerical variables create slopes.
- Interactions let offsets or slopes depend on context.

Once you see that, a lot of models are just different combinations of the same pieces.